	time axis from time = 0 to time = 50 s.
2(a)	Force constant $= \dfrac{510}{4.0 \times 10^{-3}} = 127500\ \text{N m}^{-1} \approx 1.28 \times 10^{5}\ \text{N m}^{-1}$
2(b)(i)	Work done on wire = Change in elastic potential energy in wire $= \dfrac{1}{2}(510)\left(4.0 \times 10^{-3}\right) = 1.02\ \text{J}$
2(b)(ii)	This energy is now stored as elastic potential energy in the wire.

No.	Solution
2(c)(i)	$k \propto \frac{1}{L}$ $\frac{k_{120}}{k_{1.0}} = \frac{1}{120} = 0.0083$
2(c)(ii)	$\frac{k_{120\ \text{m cable}}}{k_{120\ \text{m wire}}} = \frac{10}{1} = 10$
2(c)(iii)	The force constant of the cable is 10 times that of the single wire. For the same tension, the extension of the cable is one-tenth the extension of the single wire.
3(a)(i)	2.90